

From knowledge gaps to learning opportunities: Leveraging student questions and dual use of generative AI to support student learning at scale

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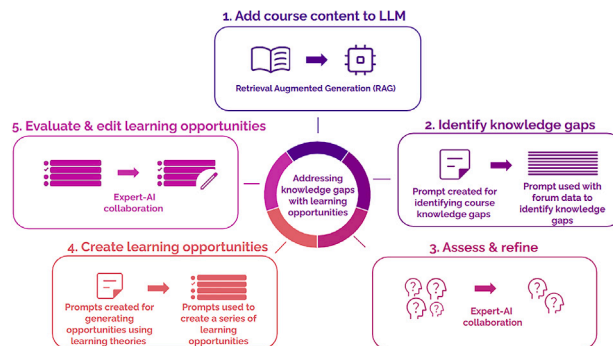
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HIGHLIGHTS

- Large classrooms are prevalent at universities, but they hinder instructors' ability to provide feedback tailored to students' needs.
- The rise of GenAI shows promise in providing automated feedback at scale; however, it often lacks the nuanced guidance of instructor-led feedback.
- Leveraging discussion forums with appropriate design and tech innovation could minimize instructor workload and address students' learning needs.
- We introduce an approach (KG2M) using LLMs and RAG to spot class-wide knowledge gaps and generate learning activities.
- Evaluation via case studies in 3 CS courses (1355 students and 2878 unique posts) and with 5 instructors emphasizes pedagogical value.

GRAPHICAL ABSTRACT



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ABSTRACT

University courses with hundreds of students have become common, particularly during early years of university studies. The sheer scale of these courses limits traditional instruction, shifting it towards a one-to-many mode of delivery. This shift reduces student–instructor interaction and tailored instructor feedback which are crucial for student success. Automated feedback systems allow scaling feedback, but they often reduce instructor contributions to student learning. This paper investigates how emerging technologies can support, rather than replace, instructors in tailoring their teaching and feedback to identify and correct student knowledge gaps at scale. To address this challenge, the paper introduces a novel technological solution: the Knowledge Gaps to Mastery (KG2M) approach. KG2M combines discussion forum data with course-specific content and leverages large language models (LLMs) and Retrieval-Augmented Generation (RAG) for the dual purpose of identifying prevalent class-level knowledge gaps and transforming them into targeted learning activities and formative assessments.

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The approach was deployed across three computer science courses with a combined enrollment of 1,355 students and evaluated through semi-structured interviews with five instructors. Results indicate that instructors found the tool intuitive and pedagogically valuable, particularly for surfacing knowledge gaps and generating actionable teaching insights. The paper reports on the tool, the evaluation, and the current limitations of the approach that emerged during instructor evaluation.

1. Introduction

The increasing scale of higher education presents a significant pedagogical challenge straining traditional models of instruction (Ryan et al., 2021). As student numbers in individual courses grow into hundreds or thousands, especially in Computer Science (CS) (Camp et al., 2017), instructors turn to pedagogies of scale (Ferguson & Sharples, 2014), such as one-to-many lectures and hybrid formats. This shift results in a reduced capacity for student-instructor interactions and a decline in tailored instructor feedback (Pardo et al., 2019). Reduction of teaching tailored to the needs of all students and diminished possibilities for instructor feedback is problematic. Instructor feedback, unlike that provided by peers or algorithms, can offer nuanced guidance that can correct deep-seated student knowledge gaps (Banihashem et al., 2024). Importantly, explanations of domain-specific issues provided by instructors show students ways of thinking foundational to their discipline (Istenič, 2021; Planar & Moya, 2016), which cannot be easily substituted by the emergent forms of student support at scale.

This paper proposes a novel technological solution to help instructors support the needs of their many students. To this end, we propose a novel use of online discussion forums in large courses. The idea that online discussions allow information exchange between students and instructors is well-accepted and widely adopted in teaching for asynchronous interactions (Loncar et al., 2014; Poquet et al., 2018), fostering flexible participation, knowledge construction (Almatrafi & Johri, 2018), and improved retention (Joksimović et al., 2018). However, the volume of forum content generated by students in large foundational courses (Reich et al., 2015) makes it challenging for instructors to efficiently reply to students (Alrajhi et al., 2020; Yang et al., 2022) or even read through all potential student difficulties.

Making use of discussion forums in new ways is under-explored (Manorat et al., 2025), yet it can address the challenge of providing nuanced instruction at scale. Large language models (LLMs) allow for processing student inquiries shared via learning technologies (Bassner et al., 2024; Lin et al., 2025) due to their capacity to handle large amounts of text. Further integration of Retrieval-Augmented Generation (RAG), a technique that combines the retrieval of relevant information from external data sources with the generative properties of LLMs, can then support processing of student inquiries in domain-specific contexts (Gao et al., 2025; Li et al., 2025). The integration of RAG with LLMs has already been explored in the context of responding to student queries in discussion forums, virtual teaching assistants, visualization dashboards, assessment and onboarding novice course tutors (Hicke et al., 2023; Lin et al., 2024; Taneja et al., 2024; Yan et al., 2024).

Although these approaches have demonstrated the potential of LLM-supported systems for teaching at scale, less attention has been given to applying them for the elicitation of students' knowledge gaps in discussion forums. Explicit identification of knowledge gaps from all students is critical if instructors want to address the emergent needs of all learners more effectively. Previous work highlights the potential of LLM-powered applications to deliver contextually relevant information to educational end-users. However, simply providing such information is not enough to enable its pedagogical benefits. Research in learning analytics, particularly learning analytics dashboards, indicates that without tools specifically designed to help end-users act on this information, it often remains unused and fails to become actionable insights for educational stakeholders (Kaliisa et al., 2024). Therefore, effectively guiding instructors, from identifying knowledge gaps to

helping them address these issues, requires a specifically tailored tool (Kohnke et al., 2025).

To support pedagogically meaningful instruction at scale, this paper proposes the Knowledge Gaps to Mastery (KG2M) approach that leverages LLMs and RAG to identify and address students' knowledge gaps in large courses while maintaining instructor oversight. Importantly, the approach is designed to complement instructors' expertise rather than replace it, offering actionable analytics that help overcome challenges of scale, limited measurements, and reliance on observational evidence from face-to-face classrooms. The approach is exemplified by KG2M, an instructor-facing tool that uses advanced prompting techniques to identify knowledge gaps from student discussion posts and assess them. The tool guides instructors through this process and takes the assessment of knowledge gaps one step further: It allows instructors to generate learning opportunities that address common knowledge gaps, with instructor input when and where necessary. This paper first explores, via a case study across three CS courses with 1,355 students, how instructors can use KG2M's instructor-facing tool to identify and address students' knowledge gaps. Following this, we empirically evaluate the KG2M approach through semi-structured interviews with five instructors from these courses. This evaluation is guided by the following **RQ: What are instructors' perspectives on KG2M's effectiveness and its pedagogical implications?** To answer this, the evaluation examines whether the approach leverages the pedagogical benefits of discussion forums for instructors and helps them address student needs. The paper concludes by discussing (i) other solutions to enable pedagogically effective instruction at scale and (ii) the implications of LLM and RAG-powered tools for supporting instructors and learning outcomes.

2. Related work

Instructional Strategies for Education at Scale. Delivering personalized and effective instruction to a large number of students remains a significant hurdle in educational settings (Davis et al., 2018; Sharples, 2016). Existing approaches have attempted to address this through various means, such as scaling instruction via video and quizzes (Cummins et al., 2015; Mirriahi et al., 2021) and improving engagement through new pedagogies of scale (Davis et al., 2018). Examples include learnersourcing, an approach that uses peer feedback to help students learn without direct instructor involvement (Khosravi et al., 2023), and gamification, which promotes motivation and engagement through game-like activities (Attali & Arieli-Attali, 2015). These solutions often fall short in addressing nuanced student questions and fostering domain-specific understanding, which typically arises from direct, instructor-mediated guidance (Wisniewski et al., 2020).

There is a critical need for technologies that support both the quality and impact of instructor-led instruction. Students must receive guidance essential for effective learning and correction of knowledge gaps. To improve quality, new technological solutions should use authentic, in-time student inquiries that reflect knowledge gaps. To improve impact, they should include specific activities that help students remedy these knowledge gaps. Since our proposed approach aims to both identify knowledge gaps and develop instructional support to address them, we now review related work on (i) identification of knowledge gaps and (ii) generation of learning opportunities.

Identifying student knowledge gaps. Early approaches to identifying students' knowledge gaps, such as quizzes with closed- or open-ended questions, often require human labeling and lack the

responsiveness needed to offer feedback when students need it most. A semi-automated method requiring minimal human labeling was suggested by Stephens-Martinez et al. (2016). This approach predicted knowledge gaps through statistical co-occurrence patterns and relied on recall and precision values (Stephens-Martinez et al., 2016). Recent advances in Knowledge Tracing (KT) show promise for real-time detection of knowledge gaps by extending models beyond binary assessments (Park et al., 2024). However, these approaches are limited to multiple-choice questions (MCQs) and require ad hoc mapping between concepts, knowledge gaps, and MCQs, which restricts applicability. Recently, Scarlatos et al. (2025) proposed an approach using LLMs to annotate tutoring dialogues, extract knowledge components, and evaluate whether learners correctly understood them. This was followed by applying established KT methods to predict the correctness of the next dialogue turn. While previous research has explored the shift from traditional assessments to data-driven approaches, *to the best of our knowledge, no prior work has utilized LLMs to derive knowledge gaps from discussion forums – a key contribution of our paper.*

Automatic generation of learning opportunities. A growing body of work demonstrates how LLMs can enhance the automatic generation of learning opportunities. For example, Denny et al. (2023) examined whether LLMs could generate educational resources comparable to those produced by computer science students. They found that LLM-generated resources were less variable in length and more consistent with existing examples. Other research has focused on creating specific types of assessment items. Jury et al. (2024) explored the use of LLMs to produce worked examples in an introductory programming course, though they noted shortcomings in the depth and structure of explanations. Similarly, Hwang et al. (2024) investigated the ability of LLMs to generate MCQs targeting different levels of Bloom's taxonomy. Results showed strong alignment between human- and AI-generated MCQs for most levels. A novel approach by Denny et al. (2024) introduced "prompt problems," a type of assessment where students craft prompts to guide LLMs in solving problems. Chan et al. (2025) investigated how different prompting methods influence quality and accuracy when leveraging GenAI to produce learning opportunities. They found the Chain-of-Prompting with code prompting technique to yield superior results compared to standard prompting, while noting high variance across STEM subjects. In contrast to fully automated generation of learning opportunities, Choi et al. (2024) suggested an advanced method that enables instructors to use LLMs to transform lecture transcripts into educational dialogues. This approach requires instructors to guide the AI through multi-step processes such as generating rubrics and questions. While LLM capabilities are promising, challenges such as hallucination – where AI produces inaccurate content – remain. RAG has been proposed as a solution, as it grounds AI outputs in verifiable data, improving accuracy and reducing errors (Gao et al., 2025; Li et al., 2025; Wang et al., 2024).

Instructor-facing Tools. Previous studies show that LLM applications can provide useful, relevant information to students and give teachers tools to create custom learning materials. Conceptually close work to ours was accomplished by Zhang et al. (2023), who developed "VizProg," a tool that supports instructors in monitoring students' coding progress in large computer science (CS) classes. This tool enables instructors to provide tailored feedback at scale. The authors evaluated the tool with instructors and found that they could identify knowledge gaps more effectively compared to conditions without tool use. However, the tool's design and usage are strictly limited to instructional settings in CS courses, such as live-coding programming exercises, and may not generalize to other educational contexts or data sources.

One of the most prolific and well-studied representations for instructor-facing tools is dashboards. These dashboards support instructors' awareness of important events in their course and allow interaction with curated representations of student data for various educational purposes. Examples include identifying students who may need closer attention (Kaliisa & Dolonen, 2023; Martinez-Maldonado, 2019),

providing personalized feedback (Pardo et al., 2018), and assessing the effectiveness of interventions and learning designs (Mangaroska & Giannakos, 2019). However, previous research on instructor-facing dashboards highlights that insights gained from these tools must translate into pedagogical actions (Amarasinghe et al., 2020; Molenaar & Knoop-van Campen, 2019) or improvements to learning design (Bakharia et al., 2016). However, taking pedagogical actions and transforming insights into actions does not happen inadvertently and requires a carefully tailored design of such tools to enable instructors to achieve this (Kaliisa et al., 2024). As such, addressing the dual challenge of identifying knowledge gaps and enabling instructors to utilize resulting insights for effective pedagogical actions requires a solution with a tailored design (Kohnke et al., 2025).

3. Approach

In this section, we present our proposed technological solution, KG2M, which addresses two key challenges: (i) identifying common knowledge gaps from online discussion forums and (ii) generating learning opportunities with instructor oversight to address these knowledge gaps. An overview of the approach taken by KG2M is shown in Fig. 1, and its architecture is presented in Fig. 2. Our approach does not introduce new methods for content creation; rather, it applies existing best practices.

The proposed approach consists of **five steps**, building upon the authors' prior work on a framework for the ethical application of generative AI to support instructional activities (Pozdniakov et al., 2024) and drawing on the approach outlined in (Choi et al., 2024). Below, we describe each step and illustrate it with reference to the instructor interface in KG2M.

Step 1: Add course content to the LLM. Building on evidence of the benefits of RAG for educational tasks (Hicke et al., 2023; Lin et al., 2024; Yan et al., 2024), we augment the underlying LLM in KG2M by incorporating auxiliary course content, including lecture notes, tutorials, and assessment tasks. Technically, this step involves creating a vector database where embedding representations of learning materials are stored. These embeddings enable integration with the RAG architecture and querying logic for content, along with prompts that guide the LLM's interaction with this content. Conceptually, once relevant course content is identified, instructors must carefully define retrieval instructions, such as criteria for querying specific learning content from the vector database.

Step 2: Identify knowledge gaps. This step develops prompt pipelines that leverage discussion forum data to uncover class-level knowledge gaps. Inputting these prompts into an LLM, then analyzes student posts individually or in batches to detect knowledge gaps. Subsequently, an LLM assists in summarizing and refining the knowledge gaps into broader categories. To ensure instructional validity and enable scrutable review of intermediate results, the prompt pipeline should align with steps defined in educational frameworks and pedagogical theories (Mollick & Mollick, 2023; Pozdniakov et al., 2024) or closely mirror processes used by disciplinary experts. This alignment allows instructors to examine intermediate outputs when necessary, enhancing transparency and trustworthiness (Felzmann et al., 2020).

While advanced prompting techniques, such as mutation prompts, emotional stimuli (e.g., engaging emotions to trigger deeper cognitive processing) (Li et al., 2023), few-shot learning, and chain-of-thought (CoT) prompting (Schulhoff et al., 2024), could improve effectiveness, research shows that their performance is highly context-dependent. For example, Almatrafi and Johri (2025) compared the classification accuracy of LLMs using different prompts for course learning outcomes. They found that a simple zero-shot prompt achieved moderate to substantial agreement with expert annotations across Bloom's taxonomy levels, while CoT did not outperform the zero-shot baseline. Similarly, Wang et al. (2023) reported that using three few-shot prompts with

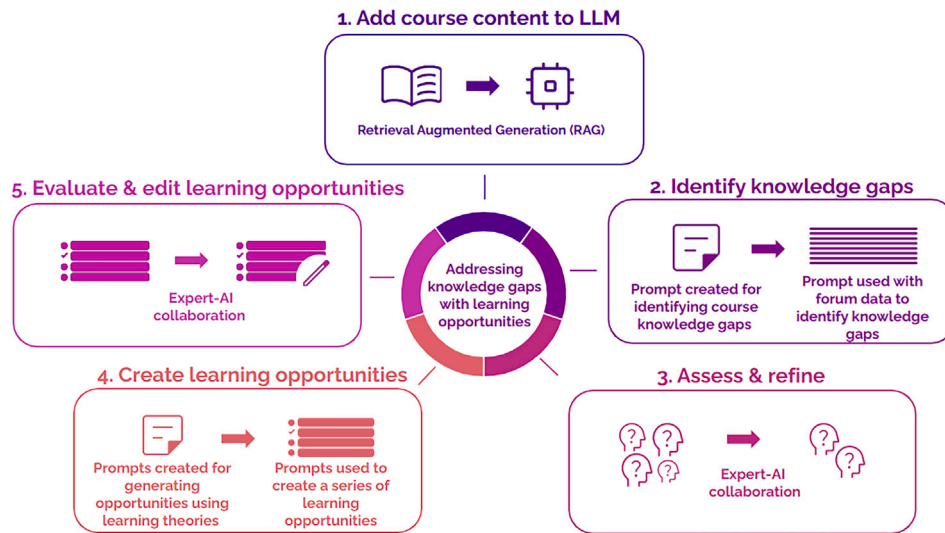


Fig. 1. Overview of the proposed approach in KG2M to discover knowledge gaps from students' discussion forums and to use them to create learning opportunities.

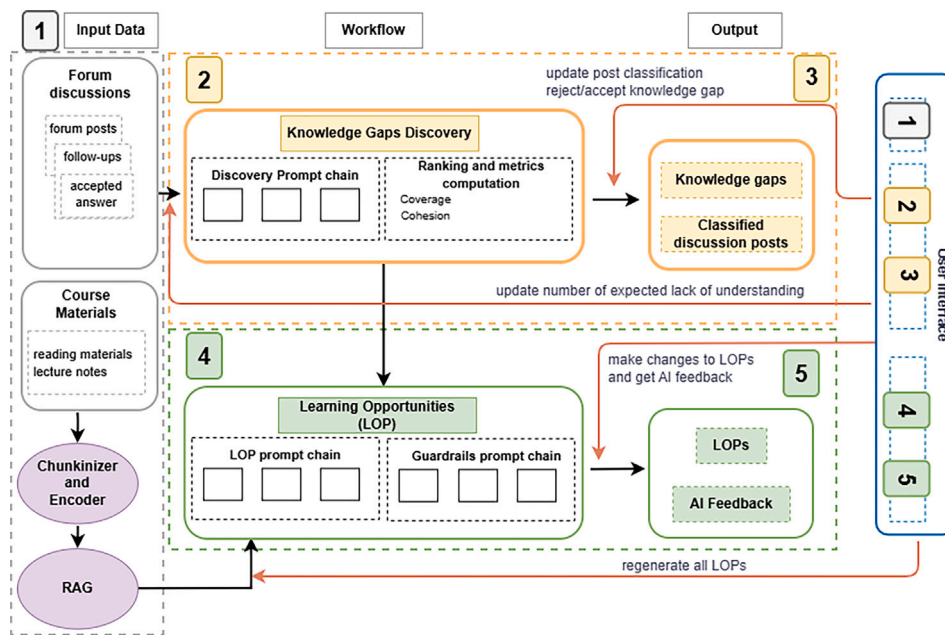


Fig. 2. Overview of the KG2M's architecture.

reasoning examples achieved greater alignment with human annotations compared to 0-, 1-, or 2-shot prompts without reasoning when classifying comments from students watching lecture recordings. These findings highlight limitations of certain techniques, such as CoT, which can lead to over-generalization, while few-shot prompting with extended context often yields satisfactory results. Alternative approaches for robust identification include agentic workflows, such as using multiple LLMs in a sequential pipeline where each model accesses previous outputs. This method reduces speculative or inaccurate responses and improves reasoning accuracy (Gosmar & Dahl, 2025). Prompting instructions for this step are shown in Listing 1, instructions 1–3.

Another key consideration is the choice of LLM, which depends on factors such as time, cost, and task complexity. Smaller, more efficient LLMs can increase response speed and reduce costs (Fawzi et al., 2024), making them ideal for real-time interactions. Conversely, reasoning-focused LLMs (Guo et al., 2025) offer greater precision and contextual

understanding but incur higher costs and slower processing speeds. These models may require asynchronous use, where instructors are notified once results are ready. Balancing these trade-offs and offering instructors options is crucial for uncovering knowledge gaps while aligning with resource constraints and pedagogical goals.

Step 3: Assess and refine the identified knowledge gaps. This step involves collaboration between instructors and AI to assess and refine knowledge gaps identified in student questions from discussion forums. To support this process, KG2M computes two key metrics in B1, namely, coverage and cohesion. *Coverage* is a number of posts related to a specific knowledge gap. *Cohesion* is a measure of how semantically consistent and tightly related the posts are to the identified knowledge gap (Fig. 4). Coverage is determined by a classification algorithm that associates each post with zero or more identified knowledge gaps (see Listing 1, instruction 4). Cohesion is calculated by obtaining embeddings of the related posts, computing their centroids, and then measuring

Algorithm 1 Identifying student knowledge gaps.

Require: Student Questions $Q = \{Q_1, Q_2, \dots, Q_M\}$

```

1:
2: // Instruction 1: Identify Knowledge Gaps (KG):
3: for question  $Q_i \in Q$  do:
4:   Use LLM to identify and describe the knowledge gap ( $KG_i$ ) in  $Q_i$ .
5:   if none then output 'No knowledge gap'.
6:   end if
7: end for
8:
9: // Instruction 2: Accumulate Knowledge Gaps ▷ Few-Shot Prompting
10: for batches  $\mathcal{KG} = \{KG_1, KG_2, \dots, KG_M\}$  of 10 do
11:   Group similar knowledge gaps in  $\mathcal{M}$ .
12:   Generalize each group into  $N$  distinct categories  $C_1, \dots, C_N$ .
13: end for
14:
15: // Instruction 3: Classify low-level knowledge gaps into class-level
▷ Few-Shot Prompting
16: for  $\mathcal{KG} = \{KG_1, KG_2, \dots, KG_M\}$  do
17:   Assign  $KG_i$  to one of the categories in  $C$ .
18: end for
19:
20: // Instruction 4: Classify Student Questions:
21: for question  $Q_i \in Q$  do
22:   Based on results from instruction 3, assign  $Q_i$  to one of the
   categories in  $C$ .
23: end for
24:
25: return Classified Questions ( $Q = \{Q_1, Q_2, \dots, Q_M\}$ ) and Knowledge
   Gap Categories  $C = \{C_1, C_2, \dots, C_N\}$ 

```

the average cosine similarity between the post embeddings and their centroids. Higher values reflect stronger cohesion, while lower values suggest weaker cohesion.

Step 4: Create targeted learning opportunities. In this step, tailored learning opportunities targeting knowledge gaps are generated using RAG with course-specific content, data on identified knowledge gaps, and associated posts and questions. Unlike the evidence regarding prompting for identification and classification described in Step 2, recent work suggests that prompting techniques such as self-refine (He-Yueya et al., 2024; Song et al., 2024) and CoT (Chan et al., 2025; Xu et al., 2025) achieve state-of-the-art results for iterative improvement of model outputs. Our approach incorporates these techniques.

Prior work emphasizes that CoT produces qualitatively superior results for creating tailored learning opportunities (Xu et al., 2025). Importantly, Xu et al. (2025) recommends combining reasoning instructions explicitly added by researchers with those implicitly generated during LLM inference. While the number of prompts depends on the desired complexity of learning opportunities, and should be considered in a cost-benefit trade-off, we achieved satisfactory results with the following steps. The instructions in CoT consist of four parts. The input to the first part is a knowledge gap description and its associated discussion forum posts. The first part is responsible for *idea brainstorming* and produces an arbitrary number of ideas for a learning opportunity to address a knowledge gap. The second part, *idea selection*, decides on the most appropriate learning opportunity idea and generates step-by-step instructions required to create the opportunity (see Algorithm 2, instructions 1–2). The third part, *learning opportunity self-refinement*, leverages RAG technique by using the knowledge gap description to retrieve relevant learning materials to align the opportunity with the expected difficulty and level of thinking (see Algorithm 2, instructions 3). This step also leverages the self-refine prompting technique, which initiates iterative improvements based on feedback from LLM (He-Yueya et al., 2024).

Algorithm 2 Generate targeted learning opportunities.

Require: Knowledge Gaps $KG_i \in \{KG_1, KG_2, \dots, KG_M\}$

```

1: Difficulty  $D_j \in \{D_{low}, D_{medium}, \dots, D_{high}\}$ 
2: ▷ CoT with guidance
3: // Instruction 1: Generate learning opportunity ideas
4: Use LLM to generate  $N$  learning opportunity ideas:  $I = \{I_1, I_2, \dots, I_N\}$ 
5: Output:  $I$  in JSON format
6:
7: // Instruction 2: Select the best idea
8: for batches  $I = \{I_1, I_2, \dots, I_k\}$  of  $k$  do
9:   Select the best idea  $I_c$ 
10:   Decide on the best idea from  $k$  batches
11: end for
12:
13: // Instruction 3: Generate and Iteratively Refine Learning Opportunity
14: Retrieve relevant context based on  $I_c$  using cosine similarity ▷ RAG
15: Initialize  $InterimResult \leftarrow \text{"None"}$ 
16: Initialize  $Refined\_LOP \leftarrow \text{None}$  ▷ Refined Learning Opportunity (LOP)
17:
18: // Iterate for refinement ( $N$  can be predefined or dynamic) ▷ Self-Refine
19: for  $i = 1, 2, \dots, N$  do
20:   Self-refinement using  $KG$ ,  $I_c$ , and  $InterimResult$  to generate  $Refined\_LOP$ 
21:   Update  $InterimResult \leftarrow Refined\_LOP$ 
22: end for
23:
24: // Instruction 4: Adjust Difficulty
25: Use LLM to adjust  $Refined\_LOP$ , accounting for  $KG$  and  $D$ 
26:
27: return  $Refined\_LOP$ 

```

Lastly, based on input from the instructors, the fourth part, *adjust difficulty of learning opportunity*, involves modifying the resource to match a specified level of difficulty or conceptual depth (see Algorithm 2, instruction 4).

Step 5: Evaluate the quality of learning opportunities and make necessary revisions. Step 5 involves collaboration between instructors and AI to evaluate and refine the quality of generated learning opportunities. This process draws on research in human-centered GenAI (Kim et al., 2024; Pozdniakov et al., 2024; Shankar et al., 2024) promoting workflows where instructors verify LLM outputs through structured “lenses”. The step uses zero-shot prompting to provide initial feedback based on predefined criteria such as correctness, contextual depth, quality of distractors (for MCQs), and alignment with the identified knowledge gaps. After reviewing GenAI feedback, instructors decide how to proceed with revisions to the learning opportunities.

4. Knowledge gaps to mastery (KG2M)

4.1. Implementation details

For the implementation, we used OpenAI’s GPT-4o mini for discovering knowledge gaps and Google’s LearnLM-1.5-Pro-Experimental for generating learning opportunities and providing evaluation feedback. As described in the previous section, our prompt design combines zero-shot prompting, CoT, and self-refinement techniques. To emphasize the role of interfaces and visual guidance while keeping human expertise central to the pedagogical process, outputs were presented through the interfaces shown in Figs. 3–5.

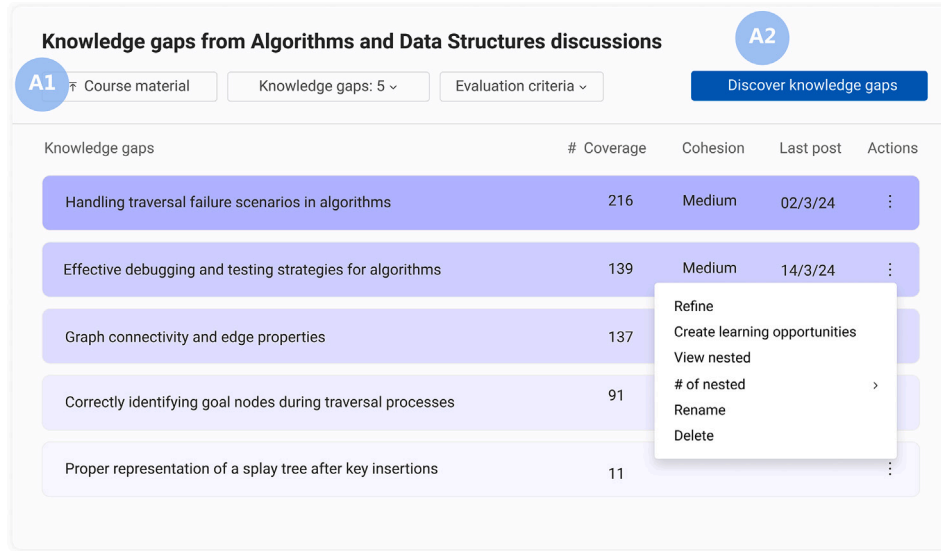


Fig. 3. KG2M's interface for Steps 1 and 2 of the process.

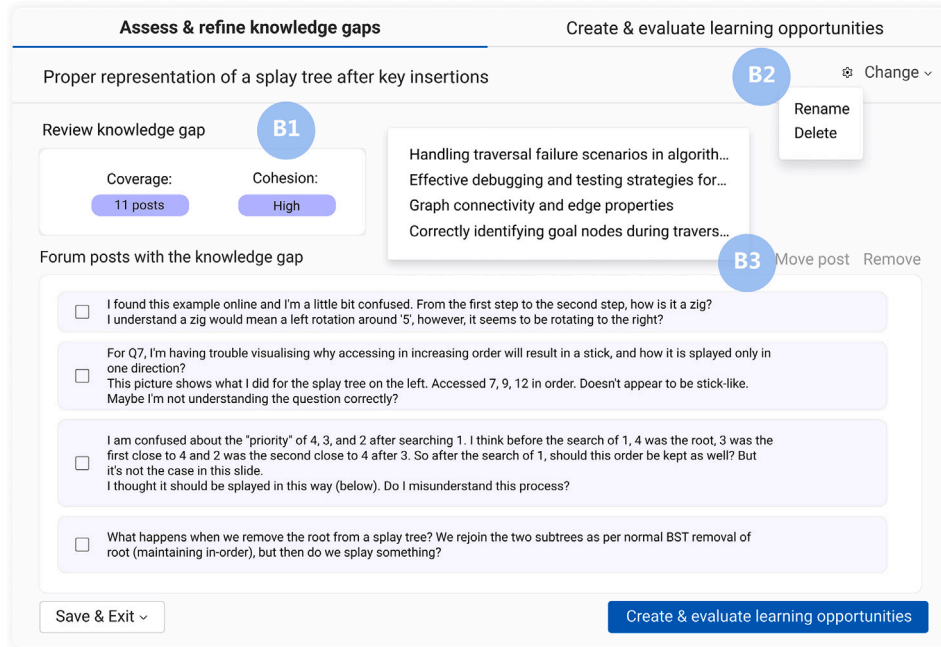


Fig. 4. KG2M's interface for Step 3 of the process.

4.2. Interface

Identify and review knowledge gaps. Fig. 3 shows the instructor interface in KG2M for Steps 1 and 2. It guides instructors through uploading relevant course material, selecting the knowledge gap discovery approach (lightweight or full, depending on the precision–cost trade-off), and specifying the maximum number of class-level knowledge gaps (A1). Instructors then click *discover knowledge gaps* (A2) and review the resulting knowledge gaps. Next, they inspect a knowledge gap by selecting among several *actions* (A3) and decide whether to refine it or create a learning opportunity targeting it, as discussed in Step 3.

Instructors would use these metrics to engage in three activities to review knowledge gaps. First, instructors could *refine the wording* of the knowledge gaps to ensure clarity and precision. Second, they could *remove knowledge gaps* that are irrelevant (B2). Finally, instructors would

review the posts from discussion forums associated with the knowledge gaps, reclassifying them if necessary into a new or existing knowledge gaps. Alternatively, instructors would remove particular posts as being misclassified and not containing any knowledge gaps in the first place (B3).

Create and oversee learning opportunities. Instructors provide input on the type of learning opportunity, expected difficulty, and targeted level of thinking (Hwang et al., 2024), e.g., information recall or knowledge application, as shown in Fig. 5 (C1). KG2M then leverages the LLM to provide initial feedback based on predefined criteria such as correctness, contextual depth, quality of distractors (for MCQs), and alignment with the identified knowledge gaps (C2). After reviewing AI feedback, instructors decide whether to regenerate the opportunity, remove it if fundamentally flawed, or manually edit it to improve its quality (C3).

Fig. 5. KG2M's interface for Steps 4 and 5 of the process.

Once finalized, instructors share the refined opportunity with students via the discussion forum or directly through KG2M.

5. Case study: deploying KG2M across three CS courses

This section presents the application of KG2M to real-life data from three undergraduate, semester-long computer science courses at The University of Queensland to demonstrate the practical benefits of our approach. We deployed KG2M in three CS courses with large student intakes. The case study illustrates how instructors can benefit from regularly reviewing top knowledge gaps using KG2M. Specifically, it focuses on two major intended uses based on metrics implemented in KG2M, such as cohesion and coverage (see Section 3, Step 3). Use case one depicts how instructors would focus on knowledge gaps with high coverage and medium cohesion. This use case represents situations where many students have broad learning gaps. Use case two illustrates how instructors would prioritize knowledge gaps with low coverage but high cohesion. This use case represents the situation where KG2M enables instructors to pinpoint subtle conceptual nuances or specific ambiguities and provide individualized support. To demonstrate the benefits of our approach, we describe how instructors could use these metrics in decision-making and show examples of learning opportunities resulting from each use case.

5.1. Learning context

“Algorithms and Data Structures” teaches fundamentals of abstract information structures in primary and secondary storage. Topics include time and space complexity analysis, data structures (stacks, queues, sets, multisets, tables), and trees and graphs. “Information Systems” covers analysis, design, and implementation of relational database technology, data modeling, SQL querying, and building small-scale information systems using relational database management systems.

Table 1

Overview of student enrollment, discussion forum posts published by students, and comments across three CS courses employed in the study.

Course name	Students enrolled	Posts published	Comments
Data structures and algorithms	574	1800	4700
Information systems	556	557	597
Web information systems	225	521	1200

“Web Information Systems” introduces Web-based systems and focuses on modern development technologies. It covers MVC frameworks, databases, HTML, CSS, server-side technologies (AJAX, Web Security, Web Search, Web Services), and current trends in web information technologies. Table 1 presents an overview of student enrollment and the number of posts and comments published in discussion forums.

5.2. Identifying knowledge gaps

Table 2 shows the top five knowledge gaps ranked by coverage (number of related posts) in descending order. These typically exhibit medium cohesion, suggesting links to diverse student questions. Conversely, the last row for each course provides an example of a knowledge gap with low coverage but high cohesion. Cohesion was calculated using the algorithm detailed in Step 3 of Section 3. Knowledge gaps with high coverage and medium cohesion are valuable for instructors because they indicate broad categories of confusion prevalent among students. In contrast, knowledge gaps with low coverage and high cohesion likely represent specific points of confusion encountered by only a few students.

Overall, instructors could consult the analytics presented in Table 2 regularly, for instance, after each lecture, assignment, or major topic. This would allow instructors to identify emerging patterns and adjust instruction for future iterations. Instructors could follow the following

Table 2

Knowledge gaps discovered from students' discussions in three courses. For demonstration purposes, the table includes the first five knowledge gaps with the largest number of posts, while the last row represents a knowledge gap with low coverage and low cohesion.

Course	Coverage	Knowledge gap	Cohesion
Data structures and algorithms	216	Knowledge gap: handling traversal failure scenarios in algorithms.	Medium
	139	Confusion regarding effective debugging and testing strategies for algorithms.	Medium
	137	Assumptions about graph connectivity and edge properties leading to incorrect interpretations.	Medium
	123	Incomplete implementation of essential functions impacting algorithm correctness.	Medium
	91	Uncertainty in correctly identifying goal nodes during traversal processes.	Medium
	...	...	...
	11	Confusion regarding proper representation of a splay tree after key insertions.	High
	146	Knowledge gap regarding the concept of cardinality in ER diagrams.	Medium
	105	Knowledge gap regarding the concept of functional dependencies in a database and its determination based on given data.	Medium
	86	Knowledge gap regarding the syntax of SQL queries and common syntax errors like missing semicolons or incorrect table/column names.	Medium
Information systems	48	Confusion about total participation in database design concepts related to relationships between entities.	Medium
	41	Uncertainty about one-to-many relationships in ER diagrams and their representation in the relational model.	Medium
	...	...	...
	13	Lack of clarity on preserving all functional dependencies while normalizing relations to 3NF.	High
	58	Assumptions and limitations in database system design without comprehensive analysis of requirements.	Medium
	37	Knowledge gap regarding concepts of table relationships and attributes in database design.	Medium
	32	Belief that replicating an existing process ensures optimal database design without considering unique requirements.	Medium
Web information systems	31	Lack of clarity regarding the implementation of advanced features in web applications.	Medium
	25	Confusion about establishing SSH connections and configuring VSCode for access to UQ Cloud Zone.	High
	...	...	...
	15	Knowledge gap regarding Nginx configuration on Windows platform.	High

diverse strategies. *Based on knowledge gaps sorted by high coverage first*, instructors could identify broad learning gaps, since high coverage indicates that a significant portion of the students are struggling with this concept. The medium cohesion would suggest that while there are variations in how students are expressing their confusion, students are all circling around a central theme. As such, medium cohesion would signal a potentially broad learning gap in the teaching materials. To alleviate this, instructors could allocate more class time to discuss the concepts that appear to be misunderstood, work through additional examples, or facilitate deeper discussions to help students solidify their understanding. *Based on knowledge gaps sorted by low coverage and high cohesion*, instructors could identify subtle nuances of the concepts included in the course that are easily missed or a very specific point of ambiguity in the materials. High cohesion and low coverage indicate that only a few students encountered this specific knowledge gap. This also could indicate that students' knowledge gaps stem from a very precise and focused issue. As such, instructors could use this analytics to inform their provision of individualized student support.

5.3. Creating learning opportunities

We present three examples of generating learning opportunities in the form of MCQs based on the knowledge gaps highlighted in Table 2. The first two examples were generated from knowledge gaps with high cohesion and low coverage, while the third example addresses a knowledge gap with high coverage and medium cohesion. Each learning opportunity is generated with AI guardrails that prompt instructors to address any emerging issues. These examples are illustrated in Fig. 6.

Example 1 was created for the “Data Structures and Algorithms” course based on a narrow, specific knowledge gap (low coverage, high cohesion; see Table 2, last row). The question asks students to perform a sequence of key insertions (8, 9, 1, 6, 3, 4, 2) into an empty splay tree and identify the correct diagram representing the resulting tree after the final splay operation on key 2. AI guardrails confirm that the correct answer (D) accurately reflects the splay tree structure. Distractors are plausible and highlight a common knowledge gap about splay tree operations, such as incorrect rotations or subtree placements. The question aligns well with the targeted knowledge gap by presenting plausible incorrect options. The key suggestion for improvement, based on AI guardrails, is to provide more explicit detail on individual steps of the splay operations to enhance contextual depth.

Example 2 addresses a knowledge gap in the “Web Information Systems” course related to a working Nginx configuration. The question describes a scenario where a web application uses Nginx as a reverse proxy, but WebSocket connections fail. Students must identify the most likely missing or incorrect Nginx configuration snippet from the given options. AI guardrails confirm that the correct answer (A) is accurate and well-supported by the explanation. Distractors are plausible, reflecting common Nginx configuration elements, which make the question challenging. While the explanation covers the necessity of specific headers for WebSockets, AI guardrails suggest improving alignment by adding deeper exploration of debugging WebSocket issues (e.g., using cURL or browser developer tools) and explaining why standard proxy configurations fail for WebSockets.

Example 3 targets a knowledge gap in functional dependencies within a database, as part of the “Information Systems” course. The question presents a real-world scenario involving customer orders and

Example 1 (DAS): Knowledge gap regarding the proper representation of a splay tree after key insertions.

Question:
Starting with an empty splay tree, perform the following sequence of key insertions: 8, 3, 9, 1, 6, 12, 4, 2. Which of the following diagrams correctly represents the resulting splay tree after the insertion of key 2? (Pay close attention to the final splay operation)

A) 2

B) 2

C) 2

D) 2

```

      /\
     1 4
    /\
   3 8
  /\
 6 9
 \
 12

```

```

      /\
     1 8
    /\
   3 9
  /\
 4 6 12

```

```

      /\
     1 3
    /\
   4 8
  /\
 6 9
 \
 12

```

```

      /\
     1 6
    /\
   3 8
  /\
 4 9
 \
 12

```

AI Feedback:
Correctness: The correct answer (D) accurately reflects the splay tree structure after the described sequence of insertions and splay operations, the diagram is correctly formatted.
Contextual Depth: It does not explicitly detail the individual steps of the splay operations (zig, zig-zag, etc.).
Distractors: The distractors are plausible and highlight common misconceptions about splay tree operations (e.g., incorrect handling of rotations or subtree placements).
Alignment: The question targets the knowledge gap effectively by presenting plausible incorrect options.

Example 3 (Infsys): Knowledge gap of the concept of functional dependencies in a database and its determination based on given data.

Question:
Consider the following table representing customer orders in an online store:
Which of the following sets of functional dependencies correctly and **minimally** represents all functional dependencies that hold in the table above? Select the most accurate and concise option.

OrderID	CustomerID	ProductID	Quantity	City	Discount
1	C1	P1	2	London	0.1
2	C2	P2	1	Paris	0
3	C1	P3	3	London	0.1
4	C3	P1	1	Madrid	0.2
5	C2	P4	2	Paris	0
6	C4	P2	1	Berlin	0.15
7	C1	P1	1	London	0.1

A) { OrderID → CustomerID,
OrderID → ProductID,
OrderID → Quantity,
CustomerID → City,
CustomerID → Discount }

C) { OrderID → CustomerID,
OrderID → ProductID,
OrderID → Quantity,
OrderID → City,
OrderID → Discount,
CustomerID → City,
CustomerID → Discount }

B) { OrderID → CustomerID,
ProductID
Quantity
City
Discount }

D) { OrderID → Quantity,
CustomerID → City,
City, CustomerID → Discount, City
→ Discount }

E) { OrderID → CustomerID,
ProductID,
OrderID → Quantity,
OrderID → City,
OrderID → Discount }

AI Feedback:
Correctness: The correct answer (a) accurately captures the functional dependencies observable in the provided dataset.
Contextual Depth: The resource lacks examples or deeper context explaining why certain dependencies hold or do not hold.
Distractors: The distractors are plausible and include common mistakes students might make.
Alignment: The resource addresses the knowledge gap by presenting a dataset and asking students to identify functional dependencies.

Example 2 (WebSys): Knowledge gap about Nginx configuration on the Windows platform

Question:
You're deploying a complex web application on a Windows Server 2019 machine using Nginx as a reverse proxy to a backend application server running on port 8080. The application uses WebSockets for real-time communication. You've configured Nginx and initially, HTTP requests are proxied correctly. However, WebSocket connections are failing with error code 1006 in the browser console. Which Nginx configuration snippet is MOST likely missing or incorrect, leading to this WebSocket connection failure?

A) `proxy_set_header Upgrade $http_upgrade;`

`proxy_set_header Connection "upgrade";`

B) `location / {
 proxy_pass http://localhost:8080;
}`

C) `events {
 worker_connections 1024;
}`

D) `http {
 include mime.types;

 default_type application/octet-stream;`

AI Feedback:
Correctness: The correct answer (A) is accurate and well-supported by the explanation
Contextual Depth: The explanation sufficiently addresses why the specific configuration is necessary for WebSocket functionality and highlights the role of the headers. However, it lacks a deeper exploration of debugging WebSocket issues (e.g., verifying the headers with tools like curl or browser developer tools).
Distractors: the distractors are plausible and reflect common Nginx configuration elements.
Alignment: It could be more aligned with the knowledge gap by exploring the broader context of debugging WebSocket issues or why standard proxy configurations fail.

Fig. 6. Examples of learning opportunities produced with the proposed approach.

asks students to identify the correct functional dependency. AI feedback confirms correctness and highlights the effective use of a practical example. Distractors are plausible, increasing the challenge, and the question aligns well with the knowledge gaps by testing students' ability to determine functional dependencies from data. The key suggestion for improvement is to add more context to enhance question's depth.

6. KG2M evaluation with instructors

To assess our approach, we held semi-structured interviews with five (P1–P5) experienced university instructors teaching three CS courses. P1, P4, and P5 taught "Information Systems"; P2 taught "Web Information Systems"; and P3 taught "Data Structures and Algorithms". Following that, we conducted a structured evaluation with course instructors, which focused on (i) instructor perspectives on the KG2M approach; (ii) their reflections on the KG2M's effectiveness; and (iii) their views on the adoption of KG2M and its impact on student learning. We now describe how we collected these responses from the instructors and their qualitative analysis and synthesis of the lessons learned from KG2M implementation.

6.1. Method

Participants. All participants who participated in the study worked at the same university. P1 had extensive experience with teaching "Information Systems", having completed ten offerings of this course, in the capacities of the course coordinator and an instructor, and had prior teaching experience. P2 previously delivered three offerings of "Web Information Systems" and had prior teaching experience. P3 was an instructor in two offerings of the "Data Structures and Algorithms" course. They were significantly involved in developing course materials, actively reworking and improving existing content to better align with teaching goals. P4 taught the "Information Systems" course ten times, serving in both capacities as an instructor and the course coordinator. They have been involved in the continuous revisions of the course and had prior experience teaching other courses. P5 previously taught the "Information Systems" course for fifteen years and had vast experience teaching other courses at the university.

Participants were asked about their teaching experience, GenAI proficiency and use of discussion forums. Participants highlighted frequent forum use to identify challenges and knowledge gaps, noting high post

volume near deadlines and first-time course offerings. The self-assessed general GenAI and pedagogical GenAI proficiency of P1–P3 was high general proficiency (with P1 also having high pedagogical GenAI proficiency), while P2 and P3 had medium pedagogical GenAI proficiency. P4 and P5 had medium general GenAI proficiency, with P4 at medium and P5 at low pedagogical GenAI proficiency.

All participants emphasized the high frequency of use and significant role of discussion forums in teaching practice, viewing forums as crucial tools for identifying student challenges and knowledge gaps, acting as a primary channel for student support and knowledge sharing. A consistent theme across all interviews is the overwhelming volume of forum posts, particularly around assessment deadlines and in situations where the course is being taught for the first time.

Procedure. The semi-structured interviews were conducted by two researchers. At first, participants were presented with an overview of the approach and a walkthrough of the tool's interface during a short onboarding session lasting approximately ten minutes. This briefing covered UI principles, navigation, and the underlying workings of the tool, with time for questions. Following onboarding, participants used the prototype to review AI-identified knowledge gaps from forum data and evaluate their authenticity and prevalence. This step typically took between twenty and thirty minutes. For the knowledge gaps that they selected, participants assessed the generated learning opportunities for quality and alignment. We relied on the implementation of the approach in three computer science courses, described previously in Section 4. To emphasize the role of interfaces and visual guidance while keeping human expertise central to the pedagogical process, the outputs were presented through the interfaces shown in Figs. 3–5. Lastly, we asked participants to critically consider the implications of KG2M tool being deployed in the classroom, accessible to both instructors and students, focusing on its impact in terms of pedagogy, ethics, and practical implementation.

Importantly, when comparing time spent with previous practice, one instructor (P2) noted that the approach enabled a structured review of knowledge gaps, something that could not be achieved otherwise. As such, a direct time comparison was not feasible, since instructors would not normally engage in systematic aggregation of knowledge gaps without the tool.

Interview Analysis. Following recommendations for qualitative evaluation (McDonald et al., 2019), two researchers took notes during the interviews, which were audio recorded and automatically transcribed using Zoom. The interview transcripts were then reviewed and discussed by two researchers with prior experience in qualitative methods, who also incorporated their individual notes and observations from the interview sessions. Following that, the researchers conducted a thematic analysis of the interview data, systematically identifying, analyzing, and reporting themes within the data. This analysis was directly guided by our research questions, with the identified themes providing the answers to each. Through iterative discussions, the researchers collaboratively grouped the responses into broad, well-defined themes, ensuring consensus and robust findings.

6.2. Results

The overall results of the evaluation are presented in Table 3.

Theme 1 – KG2M Reflections on Approach. Overall, all participants unequivocally supported the idea of identifying knowledge gaps and generating learning opportunities, with P2 describing the approach: “Even in handling all the forum posts just because of the magnitude of what comes through ... it's sometimes harder to see that higher level of what those underlying concepts might be.” P3 particularly appreciated the reduced cognitive load associated with coming up with the learning opportunities: “It takes a lot of load off of me having to come carefully with some questions.” However, P5 stressed the need to ‘close the [teaching inquiry] loop’ by evaluating the effectiveness of the learning opportunities, noting that “There should be a closing of the loop to see if those learning

Table 3
Summary of key insights.

Themes	Main insights
Theme 1. KG2M approach	–Equivocal support of the idea of identification of knowledge gaps and generation of learning opportunities as “insightful” and “actionable” (P1–P5) –An additional step could be required to evaluate the effectiveness of learning opportunities but might be difficult to implement (P5) –Instructors would benefit from a temporal presentation of knowledge gaps, such as weekly or bi-weekly filtering (P3 and P5)
Theme 2. Authenticity & Effectiveness	–Identified knowledge gaps were genuine and common among students, and learning opportunities were well-received –Adding “severity ranking” and “concept weightiness” could additionally aid instructors in assessing the criticality of knowledge gaps
Theme 3. Implications for Instruction & Learning	–KG2M could support instructors in a variety of pedagogical tasks, i.e., coordinating efforts among instructors, targeting responses, and improving course design, by aiding their awareness and enabling structural reflection on students’ knowledge gaps (P1–P5) –Instructors saw themselves as a “final filter” before presenting insights from KG2M to students (P1, P3, P5) –KG2M could impact students by reassuring them they are not alone in their struggles (P2, P4), but there is a risk of overwhelming them with the additional workload

opportunities actually helped to dispel the knowledge gaps.” P5, however, also recognized the difficulty in practically implementing this evaluation. Similarly, P1 suggested that instructors should be able to rate or approve the identified knowledge gaps to validate their accuracy.

In terms of the user interface, all participants evaluated the interface as intuitive, with P2 describing it as “very refined and straightforward”, P4 and P5 describing it as ‘very intuitive’, and P4 emphasizing “excellent presentation of the knowledge gaps”. P3 and P5 suggested that the tool’s interface would benefit from including temporal filtering capabilities. This is best illustrated by P3: “...the binary search is given to students in the first week before they learn any formal methods, and then [is given] in the last week. [KG2M] could then compare, and show that, you know, it makes a huge improvement in their knowledge.”

Theme 2 – Authenticity and Effectiveness of KG2M. In summary, all participants agreed that the identified knowledge gaps were generally authentic and reflected common student issues, as P1 stated: “I wouldn’t say any of them are made up. I would think they’re all “knowledge gaps” that happen in the course.” P4 viewed certain knowledge gaps as a positive sign indicating deeper engagement: “I would imagine it comes from reflecting on the concepts at a slightly higher level of sophistication, which is... a positive thing.” Learning opportunities in the form of MCQ were well-received, but P1, P3–P5 wanted better alignment with course materials in terms of notations used and concepts covered. This point was exemplified the best by P3: “The question is really good, except that Bellman-Ford isn’t necessarily even assessed in my course.”

Overall, the introduction of additional metrics and rankings was requested by a few participants. P3 suggested a ‘severity’ ranking. P4 proposed a metric for the concept ‘weightiness’, noting that the tool should allow instructors to specify what is important and focus on conceptual rather than technical material: “The concept of cardinality, for example. Yes, it’s important, but only a tiny part of the course...whereas something like the concept of functional dependencies. There’s a whole module that relies on this as a foundational concept.” Interestingly, P5 suggested using GenAI to automatically suggest the difficulty and the type of

learning opportunity suitable for a knowledge gap, motivating it as follows: *“There is an assumption that the instructor will pick a correct learning opportunity for that particular knowledge gap ... it might be wrong.”*

Theme 3 – Implications for Instruction and Learning. Taken together, participants indicated they would use the tool weekly to identify and address students’ knowledge gaps. P5 acknowledged KG2M’s pedagogical value in not just reducing workload but improving the quality of teaching by enabling a more targeted and efficient response to student difficulties: *“It’s not necessarily about the workload, but [about] the quality of teaching, like, you would do more for your students through this tool that you wouldn’t be able to do otherwise.”* P4 emphasized the added value of the tool for instructors teaching the course for the first time: *“I taught recently another course. Some parts of that course are my bread and butter, and some parts of it I’ve never seen before. Again, how can I know what students would struggle with? If I am myself digesting the material week-to-week, having a tool to assist me would be just excellent and really helpful.”* P2 suggested that KG2M aids instructors to reflect on the course in a structured way: *“I can see and reflect a lot more on the course now, and it’s put it in a much more structured way inside.”* Importantly, participants saw their role as deciding how exactly each knowledge gap should be resolved prior to delivering opportunities. Particularly, participants highlighted their role as deciding whether to clarify issues with students first (P2) or simply provide additional guidance to students (P3).

The tool was seen to have positive implications for instructors as it can ‘complement, not replace, their expertise’ (P1). As such, there was a consensus among participants that it is essential for instructors to act as a ‘final filter’, vetting the tool’s outputs before presenting them to students (P1, P3, P5). As P3 pointed out: *“I wouldn’t want it to be student facing the first time, not directly ... I’d like to be a ‘final filter’ before we push any of the outputs through to students.”*

Finally, instructors suggested that using KG2M in classrooms can positively impact students’ learning experience by providing targeted responses to students’ needs (P2, P5). However, there was a concern about reaching a balance between delivering learning opportunities in a manageable way, while not overburdening students with too much extra work. This point was best illustrated by P5: *“Some students are techy and really like to investigate things. This will be gold for them. On the other hand, there is a student who would be more like: ‘Give me the past exam papers, and that’s all I want to do.’”* P2 and P4 suggested that showing common knowledge gaps back to students can reassure students that they “haven’t just encountered this alone.” However, as P4 notes, it could also create a sense of hopelessness if many students struggle with the same issue.

7. Discussion and conclusion

This work addresses the problem of limited instructor feedback in large classes, which often stems from their sheer scale. To tackle this, we propose an approach that leverages discussion forums. This dual-purpose solution aims to identify students’ knowledge gaps to meet their authentic needs and simultaneously provide personalized support through tailored learning opportunities. Crucially, the approach does not seek to replace instructors’ judgment but to complement it by providing actionable insights where decision-making is difficult due to scale and incomplete evidence. Our approach is demonstrated through an instructor-facing educational tool that combines LLMs and RAG. This tool identifies common class-level knowledge gaps and generates targeted learning activities. We exemplify the tool’s use via a case study and empirically evaluate it with five instructors using real data from three computer science courses.

Preserving Instructor Agency in AI Integration. The results highlight the significant potential of GenAI applications in educational settings, particularly in large classrooms and discussion forums. This potential is realized through the specifically tailored design of our end-user tool. Pedagogically, the tool enables instructors to collaborate with AI to detect and address learning gaps that might otherwise go unnoticed.

It moves beyond simple workload optimization. Crucially, the design philosophy underpinning KG2M is grounded in human–computer interaction and human-centered learning analytics principles (Holstein et al., 2019; Shum et al., 2019), ensuring that instructor expertise remains central to the process rather than being displaced by automation. By prioritizing end-user needs and agency, the tool exemplifies a human-centric approach to AI integration in education, where technology serves as a complement to professional judgment rather than a substitute (Holstein et al., 2019). This aligns with broader calls for hybrid human–AI systems that enhance pedagogical decision-making while preserving meaningful instructor–student interactions (Cukurova, 2024). Socially, the tool supports, rather than replaces, the dialogic and collaborative nature of learning. As such, this work contributes to the emerging field of integrating LLMs into educational tools, an area with few established models to guide implementation (Belkina et al., 2025; Glassman et al., 2024; Pozdniakov et al., 2024).

While instructors see potential in using such tools for coordinating efforts, providing class-level insights to students, and assisting novice instructors in getting up to pace with the course instruction (Tan et al., 2025), concerns remain regarding unintended social comparison effects (Fleur et al., 2023) and increased student workload. Balancing AI’s role in education requires an ethically grounded (Wiese et al., 2025), human-centric approach that enhances learning without diminishing meaningful instructor–student interactions (Alfredo et al., 2024; Khosravi et al., 2022; Yan et al., 2023).

Addressing Knowledge Gaps through AI-Augmented Instruction. The results demonstrate that instructors view KG2M as an intuitive and actionable tool that effectively identifies knowledge gaps and generates learning opportunities. This aligns with prior research on learning analytics dashboards, which suggests that surfacing insights alone is insufficient. Pedagogical interventions must be embedded into the tools to help instructors act on these insights (Kaliisa et al., 2024). Our study contributes to addressing this persistent challenge by demonstrating how AI can support instructional decision-making when properly integrated into an instructor-facing interface. However, participants emphasized that instructors must remain central to the process, acting as the final filter to validate AI-generated outputs. This finding aligns with broader discussions on human–AI collaboration in education (Cukurova, 2024; Molenaar, 2022; Pozdniakov et al., 2024), emphasizing that AI should complement rather than replace instructor expertise (Kohnke et al., 2025). Future research should explore how instructors’ pedagogical strategies interact with AI-driven insights, particularly in tailoring responses to different levels of student expertise.

Refinements for Enhanced Pedagogical Utility. While instructors found the identified knowledge gaps authentic, several areas for refinement emerged. Instructors requested weekly or bi-weekly *filtering to track evolving knowledge gaps* over time. This could enhance the granularity of insights and support just-in-time teaching interventions (Beigman Klebanov et al., 2024). Another suggested improvement was the incorporation of a *severity ranking* and a concept weight metric to prioritize knowledge gaps that are foundational to student success. This aligns with prior research indicating that not all student difficulties carry equal instructional urgency (Alrajhi et al., 2020). Future iterations of KG2M should explore AI-driven classification of knowledge gaps based on pedagogical significance. Additionally, some instructors noted discrepancies in terminology and notation between AI-generated learning opportunities and course materials. Complementary to the results of (Hicke et al., 2023; Lin et al., 2024; Taneja et al., 2024; Yan et al., 2024), our evaluation emphasizes that enabling the benefits of RAG in an authentic scenario requires careful consideration of the instructional context of the problem.

Degrees of Resolving Students’ Knowledge Gaps. The design of KG2M equips instructors with analytics for surfacing and addressing knowledge gaps. It also enables instructors’ actionability (Molenaar & Knoop-van Campen, 2018) via generation of targeted learning opportunities. However, current design and implementation of KG2M’s

scaffolding strategy are primarily remedial rather than developmental. This reflects the tool's intended role: to inform instructors' decision-making rather than directly scaffold learners into their Zone of Proximal Development (Vygotsky, 1978). While this approach aligns with the immediate goal of correcting knowledge gaps at scale, it does not fully leverage more advanced instructional scaffolding strategies to guide learning (Van de Pol et al., 2010). Future research should explore how interventions can incorporate strategies that extend beyond remediation, fostering deeper conceptual development and progression within learners' Zones of Proximal Development. Such work could examine hybrid designs that combine corrective analytics with adaptive, dialogic scaffolding to support both immediate and long-term learning gains (Luckin & Holmes, 2016).

Closing the Loop: Evaluating Learning Opportunity Effectiveness. A key concern raised by instructors was the need to assess whether generated learning opportunities effectively resolve knowledge gaps. Prior research in knowledge tracing and intelligent tutoring systems has explored real-time adaptation to student performance (Alrajhi et al., 2020; Park et al., 2024; Yang et al., 2022). However, integrating such mechanisms within discussion forum analytics remains underexplored. Future research could develop longitudinal studies to track whether AI-suggested interventions lead to measurable improvements in student understanding. One potential approach is to integrate student feedback mechanisms into KG2M, allowing students to rate the usefulness of generated learning opportunities. Additionally, learning analytics techniques such as pre-post assessment comparisons could be employed to evaluate whether AI-generated content effectively supports knowledge acquisition and consolidation.

8. Limitations and future research directions

The main limitation is the absence of real-time, authentic deployment. Instructors were unable to use the tool during the continuation of the course because our approach was deployed only after the course ended and all data were collected. Consequently, without real-time observation of student outcomes, it remains unclear whether the generated learning opportunities genuinely improve learning. Another limitation of this study is the sample size, involving five instructors teaching at a single university. This might limit the generalizability of our findings. Additionally, the evaluation focused exclusively on instructors' perspectives. Considering that the ultimate goal of the tool is to benefit students, the absence of student voices regarding the perceived usefulness and effectiveness of interventions provided by the tool represents an important gap. Future work should incorporate student perspectives to inform the tool refinements and to provide a more balanced and complete evaluation. It should also explore how diverse learner needs are addressed by the approach. A co-design approach could be used to actively involve students as partners in shaping the tool design and features, ensuring their perspectives are incorporated in the design (Holstein et al., 2019; Shum et al., 2019). By engaging diverse learners in collaborative workshops or iterative feedback cycles, the approach can surface varied needs and preferences, enabling tailored solutions that enhance inclusivity and relevance of the tool (Alfredo et al., 2024; Holstein et al., 2019; Shum et al., 2019).

Future work will examine the continuous use of KG2M, focusing on instructors' interaction with the tool and its impact on both instructors and student learning outcomes. Another limitation is the alignment of AI-generated learning opportunities with course content, as instructors noted discrepancies in terminology and notations used in the course and in resulting opportunities. Addressing this issue will require introducing strong guarantees and refining the retrieval mechanisms in RAG architectures (Gao et al., 2025; Kim & Diaz, 2024; Li et al., 2025) to ensure a closer match with instructor-specific terminology and curriculum design. AI-generated knowledge gaps and opportunities currently require manual validation by instructors. While necessary for pedagogical oversight, this may limit scalability. Future iterations of KG2M could

incorporate adaptive algorithms that learn from instructor feedback, gradually reducing the need for manual review. Future implementation of the tool could also benefit from automatic prompt optimization approaches and fine-tuning of LLMs applied together, since there is nascent evidence suggesting that these approaches could achieve better performance (Soylu et al., 2024). This direction would also require constructing reliable baselines for systematic experiments with prompts for the discovery of the knowledge gaps and generation of learning opportunities. We also acknowledge that even hybrid instructor-AI identification of knowledge gaps may be inaccurate. As such, it could lead to misaligned pedagogical interventions, reducing the effectiveness of generated interventions. Future research should investigate mechanisms to validate or triangulate students' knowledge gaps, for example, by incorporating self-reported data from students. Finally, we acknowledge the potential for Hawthorne effects, as participants could alter their behaviour during the study due to being observed. Replication in authentic, longitudinal deployments will be necessary to validate findings and assess generalizability.

CRedit authorship contribution statement

Stanislav Pozdniakov: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Jonathan Brazil:** Writing – original draft, Visualization. **Oleksandra Poquet:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Stephan Krusche:** Writing – review & editing, Writing – original draft. **Santiago Berrezueta-Guzman:** Writing – review & editing, Writing – original draft. **Shazia Sadiq:** Writing – review & editing, Supervision, Project administration. **Hassan Khosravi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Ethics approval

The study was approved by an ethics committee at The University of Queensland (ID: 2024/HE001012). The paper incorporated an observational study that utilized non-identifiable historical data from students' engagement with the course, where the privacy rights of students were fully respected.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data for this article can be found online at doi:10.1016/j.caeai.2025.100509.

Data availability

The authors do not have permission to share data.

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